Semantic Epistemic Risk versus Confidence-Based Active Retrieval During Generation

Working Draft

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Abstract

Active retrieval during generation is established prior art: FLARE predicts upcoming content and retrieves when that prediction contains low-confidence tokens [1]. We ask a narrower question: is model uncertainty equivalent to epistemic risk? Using one controlled evidence source, we compare a documented FLARE-like confidence controller with transparent semantic rules that detect stale, source-dependent, or otherwise unsupported imminent commitments. The experiment measures confidence on the actual checkpoint and deliberately populates all four confidence–risk regimes. Any contribution is evidence that semantic risk catches needs missed by uncertainty at competitive cost, not the idea of generation-time retrieval itself. A developmental Qwen3-0.6B XPU pilot populates all four measured quadrants. On 12 nominal test examples, confidence OR semantic retrieval reaches 91.7% accuracy at a 91.7% retrieval rate, versus 83.3% at 83.3% for FLARE-like retrieval. The difference is one confident stale case. Because trigger and prompt policies were revised after inspecting this pilot, it is protocol-development evidence, not a held-out performance claim.

1 Research Question

When confidence and epistemic risk are measured separately, does semantic risk-triggered retrieval improve the unsupported-claim–cost frontier over FLARE-like confidence triggering, model-only generation, upfront RAG, and explicit search/tool calling?

2 Status and Claim Discipline

The source, benchmark, checkpoint confidence probe, ten conditions, typed retrieval path, traces, summaries, threshold sweeps, and XPU pilot are implemented. The pilot was used to debug the confidence objective and evidence materializer. Consequently its `test` split is an organizational split, not untouched confirmatory data. We report it to expose mechanism behavior and failure modes. A claim that semantic risk generally improves retrieval requires a frozen replication on new examples, seeds, checkpoints, and source designs.

3 Mechanism

$$Q \to y_{1:t} \to \widehat{y}_{t+1:t+k} \to \{T_{\text{conf}}, T_{\text{sem}}\} \to \text{retrieve} \to \text{evidence} \to y_{t+1:}.$$

The model need not emit a search API call. The FLARE-like condition predicts or inspects upcoming generation, computes token/model uncertainty, retrieves with the upcoming content below a fixed confidence threshold, then regenerates or continues with evidence. We report every simplification relative to canonical FLARE. The semantic condition uses only transparent lexical, syntactic, entity, temporal, freshness, and source-dependence rules.

3.1 FLARE-like implementation and simplifications

Direct FLARE temporarily generates the next sentence, retrieves if any token probability is below a threshold, formulates a query from that predicted sentence (optionally masking uncertain spans), discards the forecast, and regenerates with retrieved documents [1]. Our controller preserves that causal order but replaces each long-form sentence with one short answer span. It records every greedy token probability, triggers on the minimum, places the question and forecast in a typed request, discards the forecast when triggered, and regenerates from compact evidence. The controlled retriever uses the request's entity and attribute rather than BM25; there is one retrieval opportunity, no low-confidence masking or question-generation model, and no iterative paragraph generation. It is therefore a FLARE-like diagnostic, not a reproduction of FLARE's long-form system.

4 Single Controlled Source

The implemented source is `atlas-controlled-registry@2026.08.1`. Its records have stable IDs and validity intervals. It contains stable facts, superseded familiar facts, unfamiliar current assignments, an unavailable record, and a conflicting pair. A typed request carries entity, attribute, as-of date, forecast, candidate answer, source version, and resource bounds. Results are SUPPORTED, CONTRADICTED, UNVERIFIED, STALE, AMBIGUOUS, or CONFLICT, with record and source provenance. Retrieval never executes generated code and returns at most eight records and 512 evidence characters.

5 Trigger Design

Paper 1.5 permits only transparent, non-learned triggers:

- lexical cues such as "latest", "current", "in 2026", "highest", "most sales";
- entity/attribute patterns;
- temporal expressions;
- narrow sentence-prefix templates indicating an imminent factual value.

The pilot rules match freshness, registry, assignment, source-dependence, and year cues, with an explicit exception for "electric current." They do not contain answer values. Their narrow synthetic coverage is a limitation, not a claim of general semantic-risk understanding.

6 Measured Confidence–Risk Design

Neural uncertainty is not an evidence-need label. Examples are assigned only after measuring confidence on the evaluated checkpoint:

Confidence	Epistemic risk	Expected diagnostic behavior
High	Low	continue without retrieval
Low	Low	expose confidence-driven over-retrieval
High	High	test confident-hallucination catch rate
Low	High	both controllers should retrieve

Thresholds are fit on development data and swept at evaluation; quadrant labels must not be assigned from researcher intuition.

For the pilot, confidence is the minimum greedy probability among tokens in the forecast span. The development threshold maximizes risk-detection F1 minus 0.25 times retrieval rate; the fitted value is 0.438. The machine-readable summary also sweeps every observed test probability. High/low confidence labels are computed only after checkpoint inference. This policy was introduced after an initial pure-F1 preflight retrieved nearly always, so the present pilot is developmental and cannot validate the policy that was tuned on it.

7 Prospective and Retrospective Modes

Prospective retrieval. Interrupt before a factual value is emitted.

Retrospective verification. Allow a candidate claim, then retrieve before treating the claim as supported.

8 Epistemic State

Return compact typed status alongside evidence:

- supported;
- contradicted;
- unverified;
- stale;
- ambiguous;
- source conflict.

All records retain source identity and timestamps where available.

9 Tasks

The 20-example pilot has eight development and 12 nominal test examples. It includes familiar facts superseded only in the controlled source, unfamiliar registry values, stable facts, values explicitly supplied in the question, an “electric current” lexical distractor, conflict, and unavailable evidence. This short-answer pilot does not yet implement a credible late-emerging multi-step information need; that required task remains future work.

10 Baselines

1. LLM only.
2. Matched anti-hallucination prompt.
3. Upfront RAG with identical source access.
4. Explicit retrieval/tool calling.
5. FLARE-like confidence trigger.
6. Semantic heuristic trigger.
7. Confidence OR semantic trigger.
8. Confidence AND semantic trigger where meaningful.
9. Retrospective verification.
10. Oracle trigger and query.

11 Metrics

We report final accuracy, unsupported-claim rate, trigger and retrieval precision and recall, confident hallucination catch rate, false retrieval for low-confidence cases without evidence deficits, retrieval rate in each measured confidence/risk quadrant, uncertainty around thresholds, correction and abstention quality, evidence status calibration, tokens, calls, and latency. Plot hallucination reduction against retrieval cost with threshold sweeps and Pareto frontiers.

12 Developmental XPU Pilot

12.1 Setup

The checkpoint is Qwen3-0.6B at pinned revision `c1899de` (full hash in the manifest), run greedily with PyTorch 2.13.0+XPU on Intel integrated graphics. The source is frozen at `2026.08.1`; seed 17 is the only seed. Five distinct generation paths per example are safely shared across conditions, while condition costs account only for calls that path would make. The run contains 200 prediction rows and 200 decision traces. On the 12 nominal test examples, measured membership is one high-confidence/low-risk, three low-confidence/low-risk, one high-confidence/high-risk, and seven low-confidence/high-risk examples.

12.2 Condition results

The diagnostic separation appears in individual cases, not a significant aggregate comparison. FLARE-like retrieval fires on all three low-confidence low-risk examples and misses the one high-confidence high-risk stale fact. Semantic rules retrieve all eight high-risk examples and no low-risk example, including that confident stale fact. Confidence OR semantic therefore adds one retrieval to FLARE-like and corrects that case, raising accuracy from 10/12 to 11/12. Upfront RAG also reaches 11/12 but retrieves on every example. All evidence-aware conditions fail the unavailable-record case because this small checkpoint does not reliably follow the required abstention policy.

Table 1: Developmental pilot over 12 nominal test examples. Accuracy includes a 95% Wilson interval. Unsupported claims are errors among eight high-risk examples.

Condition	Accuracy [95% CI]	Retrieval	Unsupported
LLM only	0.083 [0.015, 0.354]	0.000	1.000
Anti-hallucination	0.250 [0.089, 0.532]	0.000	0.750
Upfront RAG	0.917 [0.646, 0.985]	1.000	0.125
Explicit retrieval	0.500 [0.254, 0.746]	0.583	0.500
FLARE-like	0.833 [0.552, 0.953]	0.833	0.250
Semantic	0.667 [0.391, 0.862]	0.667	0.125
Confidence OR semantic	0.917 [0.646, 0.985]	0.917	0.125
Confidence AND semantic	0.583 [0.320, 0.807]	0.583	0.250
Retrospective	0.167 [0.047, 0.448]	0.667	0.875
Oracle	0.667 [0.391, 0.862]	0.667	0.125

The explicit baseline emits a parsed and correct entity–attribute request for 11 of 20 total examples (seven of 12 nominal test examples); malformed or absent tags do not retrieve. Retrospective verification performs poorly because the model often preserves its draft despite contrary evidence. These are interface and evidence-use failures, not retrieval-engine failures.

12.3 Interpretation

The pilot demonstrates the hypothesized mismatch: measured uncertainty misses one confident source-dependent error and over-retrieves three low-confidence cases without evidence deficits. Combining semantic and confidence signals improves the observed frontier at the fitted threshold. This is not yet support for a general semantic-trigger claim: both effects are represented by very few examples, the semantic cues align closely with this synthetic source design, and the protocol was tuned during the same pilot.

13 Falsification

Semantic triggering fails to add value if the FLARE-like condition catches the same cases at equal or lower cost, the semantic rules duplicate uncertainty, high-confidence/high-risk cases are immaterial, false triggers erase reliability gains, or the rules effectively encode answers.

The developmental pilot does not falsify the mechanism, because semantic rules catch one case missed by FLARE-like confidence and the OR condition improves the observed accuracy–retrieval point. It also does not pass a confirmatory evidence gate: the difference is one tuned-pilot example, and upfront RAG ties its accuracy. A frozen replication that removes the confident stale case, makes semantic false positives costly, or places the OR frontier on FLARE's frontier would eliminate the proposed added value.

14 Scope

One retrieval source only. No learned trigger, no multi-source router, no symbolic reasoning, no backtracking, and no PRA/native KV dependency. The exact entity–attribute retriever intentionally removes ranking noise. The result says nothing about web-search quality, multi-source routing, or long-form retrieval.

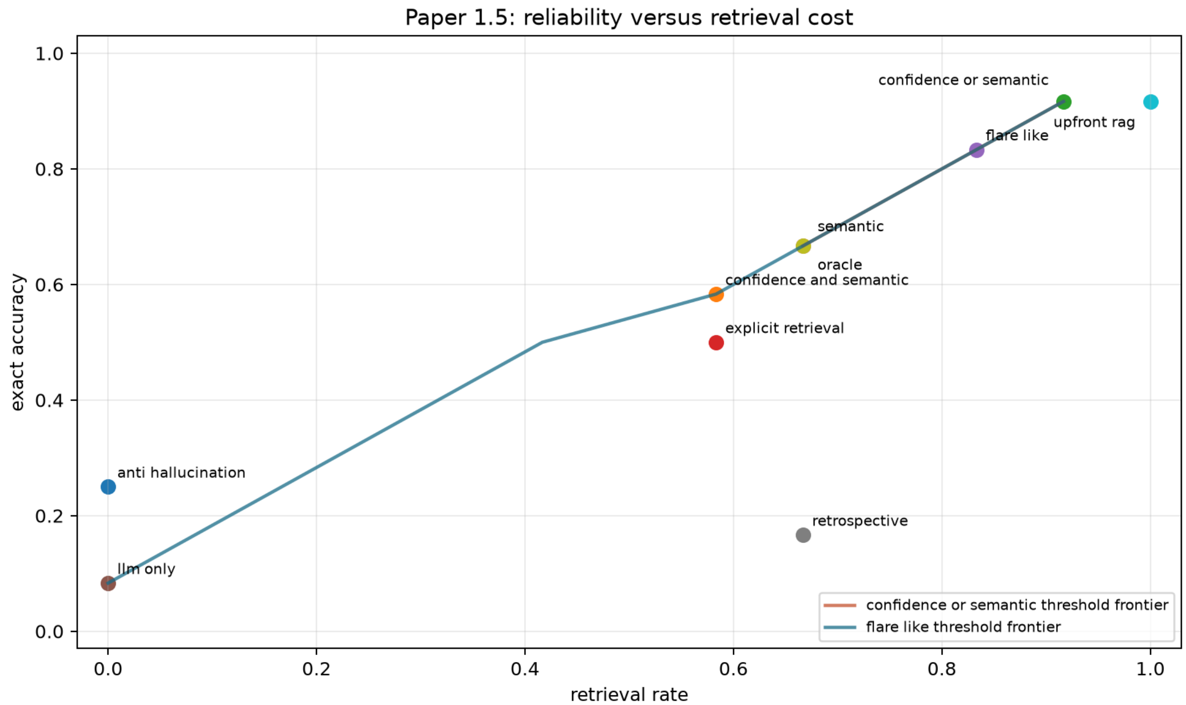

Figure 1: Exact accuracy against retrieval rate. Lines are threshold-sweep Pareto frontiers reconstructed from preserved unassisted and grounded spans. Tied points are offset only in their labels.

15 Limitations

The sample has one checkpoint, one seed, 12 nominal test examples, one high-confidence/high-risk example, and three low-confidence/low-risk examples. Protocol changes followed inspection of this same pilot. Synthetic registry cues make semantic risk easier to detect than in natural prose, while the weak checkpoint makes stable factual baselines unusually poor. The source-value materializer is compact and directive; a different evidence interface may change use and override rates. Greedy minimum-token probability is sensitive to formatting tokens and is not calibrated probability of factual correctness. The explicit textual tool baseline is not native function calling. Short answer spans omit repeated retrieval, late-emerging needs, ranking errors, and evidence synthesis. Latencies reflect full-prefix generation and are not production estimates.

16 Reproduction

Paper-specific code is under `src/ccpu/paper1_5`; artifact, metric, and device-selection utilities are reused from `src/ccpu/common` and Paper 1's backend support. From the repository root:

```
python -m ccpu paper1.5 validate \
  --config configs/paper1_5/xpu_pilot.json
$py = 'C:\Users\j.simao\.venvs\modal-llm-xpu\Scripts\python.exe'
$env:PYTHONPATH = (Resolve-Path src).Path
& $py -m ccpu paper1.5 run-hf \
  --config configs/paper1_5/xpu_pilot.json \
```

```
  --output-dir artifacts/paper1_5/xpu_pilot
python -m ccpu paper1.5 plot \
  --summary artifacts/paper1_5/xpu_pilot/summary.json \
  --output docs/papers/paper1_5/paper1_5_xpu_pilot.png
```

The manifest hashes the config, benchmark snapshot, predictions, traces, and summary and records the source version, fitted threshold, model revision, environment, repository state, and device. All 200 rows record XPU execution.

17 Conclusion

This controlled pilot separates confidence from epistemic risk on the actual checkpoint and implements a FLARE-like forward forecast alongside transparent semantic rules. The observed OR condition catches one confident stale error that confidence misses, while confidence alone over-retrieves all three low-confidence no-deficit cases. These are useful diagnostics, not a general result. Paper 1.5 now supports a frozen, larger replication; it does not yet justify a claim of superior active retrieval or progression to multiple sources.

References

[1] Z. Jiang et al. Active retrieval augmented generation. *EMNLP*, 2023. https://arxiv.org/abs/2305.06983.